

Jeongmin Lee

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Ph.D., Robotics

My research focuses on developing intelligence for robotic manipulation involving contact interactions. During my Ph.D., I worked on theories of contact modeling and solvers for high-performance robotic simulators, as well as efficient algorithms for planning, estimation, and control related to contact dynamics. Currently, I am broadening my scope to explore the intersection of contact modeling, solvers, and machine learning. Additionally, I am actively involved in creating a software suite for robotic manipulation applications.

EDUCATION

Seoul National University <i>Seoul, S. Korea</i> M.S.-Ph.D. in Mechanical Engineering	Mar. 2019 - Aug. 2024
Seoul National University <i>Seoul, S. Korea</i> B.S. in Mechanical and Aerospace Engineering	Mar. 2015 - Feb. 2019

EXPERIENCE

Senior Research Engineer, Holiday Robotics Research on physics simulation and humanoid manipulation	Mar. 2025 -
Postdoctoral Researcher, SNU INRoL Mechanics & Control group leading	Sep. 2024 - Feb. 2025
Graduate Research Assistant, SNU INRoL - Advisor: Dongjun Lee - Research on developing high performance robot simulation and contact manipulation	Mar. 2019 - Aug. 2024
Research Intern, Korea Institute of Science and Technology (KIST) Developed a Lidar localization algorithm based on CNN and ICP	May 2018 - Dec. 2018
Teaching Assistant - Control System 1 (M2794.005300) - Mechanical System Modeling and Control (M2794.013500)	Spring 2021 Spring 2020

AWARDS AND HONORS

RSS Pioneers 2025 Title: Enhancing Physics Simulators for Reliable and Scalable Data Generation (Link)
Outstanding Doctoral Thesis Award 2024 (SNU, KSME) Title: Efficient and Scalable Methods for Contact Manipulation: From Simulation to Estimation and Planning
Runner-up Award of the Embracing Contacts Workshop in ICRA 2023 A differentiable geometry & optimization algorithm for state estimation during contact
Best Manipulation Paper Award Finalist in ICRA 2021 Flexible cable simulation algorithm and sim-to-real verification
Outstanding BS Thesis Presentation Award Localization based on Lidar image using CNN and ICP

PUBLICATIONS

Journal Articles

- Jeongmin Lee**, Minji Lee, Sunkyung Park, Jinhee Yun, and Dongjun Lee, Variations of Augmented Lagrangian for Robotic Multi-Contact Simulation, IEEE Transactions on Robotics 2025.
- Jeongseob Lee, Doyoon Kong, Hojun Cha, **Jeongmin Lee**, Dongseok Ryu, Hocheol Shin, Dongjun Lee, Wrench Control of Dual-Arm Robot on Flexible Base with Supporting Contact Surface, IEEE Transactions on Robotics 2025.
- Minji Lee, **Jeongmin Lee**, Dongjun Lee, Narrow Passage Path Planning via Homotopy-Preserving Collision Constraint Interpolation, IEEE Transactions on Robotics 2025.
- Jeongmin Lee**, Minji Lee, and Dongjun Lee, Large-Dimensional Multibody Dynamics Simulation using Contact Nodalization and Diagonalization, IEEE Transactions on Robotics 2022.

Conference Proceedings

- Heeseong Lee, Sungpyo Sagong, Minhyeong Lee, **Jeongmin Lee**, Dongjun Lee, Efficient Reinforcement Learning for Sparse-Reward Autonomous Car Racing using Imperfect Demonstrations, IROS 2025.
- Sunkyung Park, **Jeongmin Lee** and Dongjun Lee, Shape Abstraction via Marching Differentiable Support Functions, CVPR 2025 (Highlight).
- **Jeongmin Lee**, Minji Lee, Sunkyung Park and Dongjun Lee, Efficient Gradient-Based Inference for Manipulation Planning in Contact Factor Graphs, IEEE ICRA 2025.
- Minji Lee, **Jeongmin Lee**, and Dongjun Lee, Narrow Passage Path Planning using Collision Constraint Interpolation, IEEE ICRA 2025.
- Harim Ji, Hyunsu Kim, **Jeongmin Lee**, Somang Lee, Seoki An, Jinuk Heo, Youngseon Lee, Yongseok Lee, and Dongjun Lee, GPU-Accelerated Subsystem-Based ADMM for Large-Scale Interactive Simulation, IEEE ICRA 2025.
- Seoki An, Somang Lee, **Jeongmin Lee**, Sunkyung Park, Dongjun Lee, Collision Detection between Smooth Convex Bodies via Riemannian Optimization Framework, IEEE/RSJ IROS 2024.
- **Jeongmin Lee***, Minji Lee*, and Dongjun Lee, Uncertain Pose Estimation during Contact Tasks using Differentiable Contact Features, RSS 2023.
- **Jeongmin Lee**, Minji Lee, and Dongjun Lee, Modular and Parallelizable Multibody Physics Simulation via Subsystem-Based ADMM, IEEE ICRA 2023.
- Minji Lee, **Jeongmin Lee**, and Dongjun Lee, Differentiable Dynamics Simulation Using Invariant Contact Mapping and Damped Contact Force, IEEE ICRA 2023.
- Minji Lee, **Jeongmin Lee**, Jaemin Yoon, and Dongjun Lee, Real-Time Physically-Accurate Simulation of Robotic Snap Connection Process, IEEE/RSJ IROS 2021.
- **Jeongmin Lee**, Minji Lee, Jaemin Yoon, and Dongjun Lee, A Parallelized Iterative Algorithm for Real-Time Simulation of Long Flexible Cable Manipulation, IEEE ICRA 2021. (**Best Manipulation Paper Award Finalist**)

Under Review/In Preparation

- **Jeongmin Lee**, Minji Lee, and Dongjun Lee, Differentiable Support Functions: Theory and Application to Contact State Estimation during Manipulation, In Preparation.

Workshops

- Minji Lee*, **Jeongmin Lee***, and Dongjun Lee, Assembly Path Planning via Variable Lifting and Physics Simulation, Robot Assembly Workshop in RSS 2023
- **Jeongmin Lee***, Minji Lee*, and Dongjun Lee, A Differentiable Formulation for Uncertain Pose Estimation during Contact, Embracing Contacts Workshop in ICRA 2023. (**Runner-up Award**)
- Minji Lee, **Jeongmin Lee**, and Dongjun Lee, Interactive Real-time Simulation of Robotic Snap Connection Process, Representing and Manipulating Deformable Objects in ICRA 2021.

GOVERNMENT AND INDUSTRIAL PROJECTS

Development of a Digital Twin Platform for Precision Assembly Tasks	Apr. 2024 - Feb. 2025
• Ministry of Trade, Industry and Energy (MOTIE) • Developing precise geometric and physical models and solvers for contact interactions	
Virtual Environment and Contact Manipulation Module for Nuclear Power Plants	Mar. 2022 - Feb. 2025
• Ministry of Science and ICT (MSIT) • Developing motion planning algorithms through contact for different emergency tasks	
Autonomous Dish Placing Framework using Robot Manipulator	Jan. 2022 - Oct. 2022
• Samsung Research • Designed dish motion planning and state estimation module utilizing sensor data	
High Speed & Accurate Simulator for Robot Manipulation	Jan. 2020 - Dec. 2021
• Ministry of Trade, Industry and Energy (MOTIE) • Developed contact solver and sim-to-real framework of deformable object manipulation	
Development of Remote Control Technology for Nuclear Power Plant Maintenance	Jan. 2019 - Dec. 2019
• Ministry of Trade, Industry and Energy (MOTIE) • Implemented on-board state estimation module for a dual-arm robot working at heights	

ACTIVITIES

Consumer Electronics Show (CES) Las Vegas, USA	Jan. 2022
• Developed and exhibited a virtual environment platform integrated with haptic glove and physics simulation (Link)	

SKILLS

Programming	C/C++, MATLAB, Python
Tools	Simulation(MuJoCo, Bullet, Isaac Sim), Optimization(NLopt, Ipopt, Knitro, Gurobi), Robotics(ROS, Drake), Learning(PyTorch, Tensorflow), Warp, OpenGL, Solidworks, LaTeX
Robots	Franka Panda, KUKA iiwa